Prabhav Rao

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GitHub | Portfolio | U.S. Permanent Resident

Education

- **University of California, Los Angeles** *June 2027*
  Bachelor of Arts in Computer Science and Linguistics Los Angeles, CA
  - Relevant Coursework: Introduction to Artificial Intelligence, Algorithms & Complexity, Theory of Computing, Statistics

Experience

- **Oracle** *June 2026 – September 2026*
  Software Engineer Intern Redwood City, CA
  - Cut 22-step onboarding from **1–2 weeks to 2–3 days** with an **AI skill** automating local setup and validating **Docker/Kubernetes**.
  - Cut JVM library delivery from **2–3 days to a few hours** with an AI skill recursively resolving **Maven/Gradle** dependency trees.
  - Verified JVM builds with unit/integration and downstream tests, then published artifacts to Artifact Hub through **Python Scripts**.
  - Prevented ID tampering in **PHP/Java REST APIs** with POST/PUT validation and integration tests, deployed through GitLab CI/CD.
  - Reduced false on-call alerts by correcting **HTTP error handling** and testing affected routes with curl and PHP test suites.
  - Enabled 2.4B-point awards by migrating Protobuf fields and Java types from **32-bit to 64-bit** across services without overflow.
  - Preserved compatibility during PHP/Java Protobuf and protoc upgrades with **3 RabbitMQ/HTTP flows** and Oracle Database checks.
- **Teens 4 Teens** *June 2025 – August 2025*
  Software Engineer Intern Charlotte, NC
  - Built a SaaS-based **applicant tracking system** onboarding **20+ HR users and 200+ applicants**
  - Engineered a scalable **Flask backend with Celery workflows** to automate resume parsing, reducing HR effort by 70
  - Implemented a **keyword-based scoring** feature to automate resume ranking and improve screening accuracy
  - Set up **CI/CD pipelines** for deployment on Vercel and Railway
  - Designed an evaluation dashboard and application website with **secure authentication and role-based access control**
- **Stanford University** *September 2024 – December 2024*
  Undergraduate Researcher Palo Alto, CA
  - Improved CPU efficiency by **10–15%** in benchmarked simulations through pipeline and cache optimization
  - Conducted 20+ controlled experiments on **instruction-level parallelism** to measure throughput and latency
  - Optimized a simplified CPU design to reduce **cycles per instruction** and boost execution performance
  - Analyzed **pipeline and cache trade-offs** in computer architecture with a postgraduate mentor
- **Kahoot!** *May 2024 – July 2024*
  Software Engineer Intern Boston, MA
  - Engineered backend features for **grade and subject categorization**, simplifying teacher workflows and reducing setup friction
  - Prototyped a **generative AI** feature to create Sparks from teacher prompts, integrating built-in **moderation and safety controls**
  - Redesigned Kahoot Sparks by conducting **50+ user prototype tests** with students and teachers to identify usability gaps

Projects

- **Multithreaded Web Server with Thread Pool** *September 2025*
  Tools: C++, POSIX Sockets
  - Developed a lightweight C++ HTTP server with a **custom thread pool**, achieving a **4× throughput improvement** under benchmarks
- **GPU-Optimized Deep Learning Library Extension** *September 2025*
  Tools: C++, CUDA, PyTorch, cuBLAS, cuDNN
  - Achieved **5× CNN training throughput** versus PyTorch with a C++/CUDA extension using cuBLAS/cuDNN matrix/convolution kernels.
- **GPU-Accelerated Ray Tracer with Deep Learning Denoiser** *August 2025*
  Tools: C++, CUDA, OpenMP, PyTorch
  - Achieved **20× faster rendering** than CPU baselines with a C++/CUDA ray tracer using BVH acceleration and optimized kernels.
  - Reduced rendering noise by **70%** using a PyTorch denoiser integrated into the GPU ray-tracing pipeline.
- **Trading Strategy Backtester** *July 2025*
  Tools: Pandas, Matplotlib, Alpha Vantage API
  - Enabled historical validation with a Python backtester implementing **SMA, RSI, and Bollinger Bands** strategies.
  - Enabled custom backtests with **Streamlit**, API/CSV data, and Sharpe, Sortino, drawdown, Calmar, and Profit Factor metrics.

Skills

- **Programming Languages:** C++, C, Java, Python, JavaScript, SQL, Go, HTML/CSS
- **Technologies & Tools:** React.js, Node.js, Docker, Kubernetes, Git, GitLab, Maven, Gradle, IntelliJ
- **Concepts:** REST APIs, Multi-threaded Programming, Protobuf, Linux/Unix, TCP/IP
- **Certifications:** Oracle Cloud Infrastructure Certified Generative AI Professional